
HK IMO Training 2020-2021
Problem Set 3
For 31 October 2020



Problem 9. Let ABC be an acute, non-isosceles triangle. The altitudes BD and CE intersect at H . Let G be the centroid of $\triangle HBC$. If the line AG bisects the segment DE , find $\angle A$.

Problem 10. Let O and I be the circumcentre and incentre of $\triangle ABC$ respectively. The A -excircle of $\triangle ABC$ touches the lines BC, CA, AB at D, E, F respectively. If the midpoint of EF lies on the circumcircle of $\triangle ABC$, prove that O, I, D are collinear.

Problem 11. In $\triangle ABC$, $\angle A = 120^\circ$. Let M be the midpoint of BC . Points D and E are chosen on the sides AB and AC respectively, such that $DE \parallel BC$. Prove that $MD + ME \geq MB$.

Problem 12. Let ω be the incircle of a quadrilateral $ABCD$. The rays AB and DC meet at point E . The rays AD and BC meet at point F . The lines AC and EF intersect at point P . Let Q be the point on ω which is closest to the line EF . Prove that the line PQ passes through the incentre I of $\triangle CEF$.